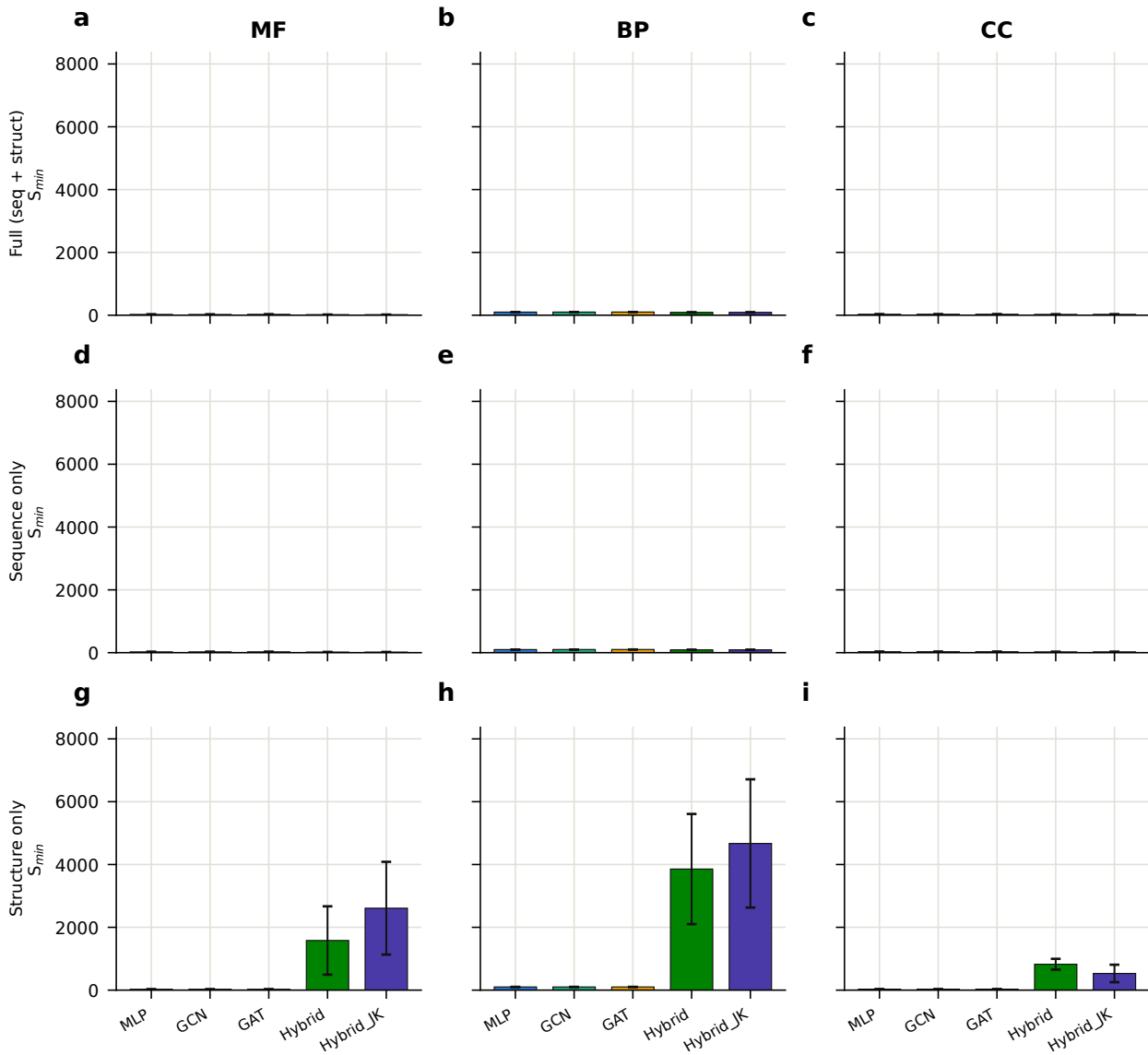




Rows are input modalities and columns are ontologies. Bars show the mean with mean $\pm$ SD across five seeds; lower is better. Bars are grouped/faceted, because model scores are not additive. F_{max}/S_{min} favour graph-aware Hybrid/Hybrid_JK. Under structure-only, MLP/GAT/GCN reach exactly the predict-nothing S_{min} (the test set's total IC), while Hybrid/Hybrid_JK saturate above the 0.99 threshold-scan bound and predict everything; both are degenerate, not tuned trade-offs.